

Generation Interconnection Combined Feasibility / System Impact Study Report

For Queue Project Q17-0903

Sabanera 230 kV
August 23, 2019

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q17-0903. The Interconnection Customer proposes a 1143.5 MW natural gas combined cycle facility interconnecting at the Sabanera 230 kV substation in the Indiana area. The request was received on December 15, 2017. The study evaluated system intact and single contingency conditions using the regional planning case, including 301 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 25 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$97,801,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
HENRY COUNTY 500-230 kV Transformer Bank	E METHUEN / LONETREE 500kV line	134.5	894	23.3
LONETREE 500/230 kV Transformer Bank	EAST METHUEN / LONETREE 500kV	115.8	811	7.2
BARAGA VLG / GRAY'S FLAT 230kV	MADELINE / E METHUEN 230kV	115.5	1,020	13.7
BARAGA VLG / MADELINE 230kV Line	E METHUEN / GRAY'S FLAT 230kV	114.9	637	10.1
15TH AVENUE S / LUCENT 230kV Line	OMNI-HACKBERRY / NORWICH 500kV	114.3	616	6.2
BURRIS PIT / HEBO 230kV Line	TREGO / BEAVER 500kV	114.2	1,021	11.3
15TH AVENUE S / NORWICH 230kV line	NORWICH / OMNI-HACKBERRY 500kV	114.1	630	6.2
JAMES DE YOUNG / BURRIS PIT 230kV	TREGO / BEAVER 500kV Line	113.5	1,046	11.3
E METHUEN / GRAY'S FLT 230kV	E METHUEN / MADELINE 230kV line	113.4	1,052	13.7
JAMES DE YOUNG / CARLEY ROAD 230kV L	E METHUEN 500-230 kV TX	110.4	781	11.0
POST ROCK / NEWPORT TIE 230kV	TREGO / BEAVER 500kV L	110.0	1,766	11.6
POST ROCK / MYRTLE PT 230kV line	TREGO / BEAVER 500kV	109.3	1,820	11.2
ONANCOCK DIESEL GENERATION FACILITY / BEAVER 230kV	BEAVER / EUFAULA 500kV Line	108.3	1,730	9.5
TROY ENERGY LLC / RICHLAND 500kV	SWAN VALLEY / TREGO 500kV	107.1	3,510	11.0
CORRIDOR / LARSON STREET 230kV	BEAVER / TREGO 500kV	107.1	851	6.4

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
MASON ST 500/230 kV Transformer Bank	BEAVER / TREGO 500kV	107.0	1,705	7.0
JAMES DE YOUNG / ADIRONDACK 230kV	ST. LAWRENCE / JAMES DE YOUNG 230kV	107.0	952	10.1
TRUMAN / LOOKINGGLASS 230kV	TREGO / BEAVER 500kV	106.5	911	6.4
BARAGA VLG 500-230 kV TX	TREGO / BEAVER 500kV	106.2	1,240	17.9
EAST METHUEN 500 kV/230 kV XFMR	TREGO / BEAVER SUBSTATION 500kV L	106.0	2,119	21.7
ST. LAWRENCE / JAMES DE YOUNG 230kV	E METHUEN / LONETREE 500kV	105.8	1,287	19.7
GALION GENERATING STA 500 kV/230 kV Transformer Bank	TREGO / BEAVER 500kV Line	105.6	1,417	8.1
ADAMS / NEW DEPARTURE 230kV Line	TREGO / BEAVER 500kV Line	105.5	1,129	5.2
IMPERIAL / EUFAULA 500kV line	TREGO / GALION GENERATING STATION 500kV	105.3	3,359	22.4
LONETREE / SABANERA 500kV	MUSKOGEE / MONTPELIER ELECTRIC GENERATING STA 500kV	105.1	3,482	42.9

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
HENRY COUNTY 500-230 kV Transformer Bank	Replace transformer with higher-rated unit	\$33,262,000	24% (\$7,857,000)
LONETREE 500/230 kV Transformer Bank	Replace transformer with higher-rated unit	\$16,829,000	21% (\$3,559,000)
BARAGA VLG / GRAY'S FLAT 230kV	Rebuild line with higher-capacity conductor	\$25,779,000	14% (\$3,477,000)
BARAGA VLG / MADELINE 230kV Line	Rebuild line with higher-capacity conductor	\$13,656,000	13% (\$1,831,000)
15TH AVENUE SOUTH / LUENT 230kV line	Rebuild line with higher-capacity conductor	\$16,996,000	13% (\$2,188,000)
BURRIS PIT / HEBO 230kV Line	Rebuild line with higher-capacity conductor	\$28,857,000	16% (\$4,693,000)
15TH AVENUE S / NORWICH 230kV line	Rebuild line with higher-capacity conductor	\$11,043,000	13% (\$1,422,000)
BURRIS PIT / JAMES DE YOUNG 230kV L	Rebuild line with higher-capacity conductor	\$57,640,000	16% (\$9,374,000)
GRAY'S FLAT / EAST METHUEN 230kV Line	Rebuild line with higher-capacity conductor	\$34,003,000	14% (\$4,586,000)
JAMES DE YOUNG / CARLEY ROAD 230kV L	Rebuild line with higher-capacity conductor	\$29,317,000	13% (\$3,862,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
NEWPORT TIE / POST OCK 230kV L	Rebuild line with higher-capacity conductor	\$26,403,000	13% (\$3,397,000)
MYRTLE POINT / POST ROCK 230kV	Reconductor line	\$10,817,000	13% (\$1,393,000)
ONANCOCK DIESEL GENERATION FACILITY / BEAVER 230kV	Reconductor line	\$10,001,000	13% (\$1,310,000)
TROY ENERGY LLC / RICHLAND 500kV	Reconductor line	\$79,709,000	8% (\$6,450,000)
CORIDOR / LARSON ST 230kV	Reconductor line	\$5,031,000	11% (\$568,000)
MASON ST 500/230 kV Transformer Bank	Replace transformer with higher-rated unit	\$35,702,000	11% (\$3,970,000)
JAMES DE YOUNG / ADIRONDACK 230kV	Reconductor line	\$11,587,000	15% (\$1,738,000)
TRUMAN / LOOKINGGLASS 230kV	Reconductor line	\$11,126,000	11% (\$1,257,000)
BARAGA VLG 500-230 kV Auto	Replace transformer with higher-rated unit	\$27,176,000	13% (\$3,649,000)
EAST METHUEN 500/230kV Transformer	Replace transformer with higher-rated unit	\$23,831,000	14% (\$3,210,000)
ST. LAWRENCE / JAMES DE YOUNG 230kV	Reconductor line	\$23,765,000	22% (\$5,218,000)
GALION GENERATING STA 500 kV/230 kV Transformer Bank	Replace transformer with higher-rated unit	\$44,126,000	12% (\$5,412,000)
ADAMS / NEW DEPARTURE 230kV Line	Reconductor line	\$7,262,000	12% (\$853,000)
IMPERIAL / EUFAULA 500kV line	Reconductor line	\$53,970,000	12% (\$6,609,000)
SABANERA / LONETREE 500kV	Reconductor line	\$48,856,000	20% (\$9,918,000)
	Total allocated to project		\$97,801,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.